

To be done in groups of 3.



جامعة الامير سلطان
PRINCE SULTAN
UNIVERSITY

Project: Solving and Analyzing Linear Systems in Python

Objective:

Solve a system of linear equations using three different methods (with at least one method not relying on Python libraries and one iterative method), compare their performance, and apply them to a real-world problem.

Project Requirements

1. Solve a System

- Implement three methods to solve a system of linear equations.

2. Performance Comparison

- Create large random systems (e.g., 100×100) and solve them using the three methods.
- **Compare** the methods by:
 - **Execution time** (using time or timeit)
 - **Accuracy** (compare solutions)
 - **Iteration count** for the iterative method

3. Real-World Problem

- Choose a real-world problem (e.g., electrical circuits, chemical equations, economic models, or structural analysis) and model it as a system of linear equations.
 - Solve it using all three methods.
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Deliverables (Due by November 17th)

1. Python Code

Submit a .py file or Jupyter Notebook with your code, including:

- Method implementations
- Performance comparison
- Real-world application solution

2. Report (Max 2 Pages)

Submit a brief report that includes:

- **Methods Explanation:** Briefly describe each method.
- **Comparison of Methods:** Compare execution times, accuracy, and iteration counts.
- **Real-World Problem:** Explain your chosen problem, the system formulation, and the results.
- **References**